

1. b) Data & pointer to the next node
2. c) Null
3. b) Self-referential structure.
4. b) $head == Null$
5. b) $newnode \rightarrow next = head$; $head = newnode$.
6. b) $temp = head$; $head = head \rightarrow next$; $free(temp)$;
7. c) Traversing from tail to head.
8. b) n.
9. c) Copying the data of $P \rightarrow next$ into P , then deleting $P \rightarrow next$.
10. b) while ($P != Null$).
11. b) $P \rightarrow next == Null$.
12. c) Two pointers advancing at different rates.
13. c) Three.
14. b) Exactly one node.
15. b) $P = (struct node *) malloc (Size of (struct node))$
16. b) Avoid special-case handling for insertion & deletion at the head.
17. c) The first node
18. b) It does not support random (indexed) access.
19. b) Loss of access to all nodes, causing a memory leak
20. b) A slow pointer & a fast pointer.
21. c) The last node.
22. b) Nodes may be scattered anywhere in memory
23.

```
struct node {  
    int data;  
    struct node * next;  
};
```
24. 30 50 70
25. Connect the insertEnd() function.

```
void insertEnd (struct node **head, int x) {  
    struct node *t = (struct node *) malloc (Size of (struct node));  
    t->data = x;  
    t->next = Null;  
    struct node *p = *head;
```

```
while (p->next != NULL)
```

```
p = p->next;
```

```
p->next = f;
```

```
}
```

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26) Struct node* reverse (struct node* head) {

```
    struct node* prev = NULL;
```

```
    struct node* curr = head;
```

```
    struct node* next = NULL;
```

```
    while (curr != NULL) {
```

```
        next = curr->next;
```

```
        curr->next = prev;
```

```
        prev = curr;
```

```
        curr = next;
```

```
    }
```

```
    return prev;
```

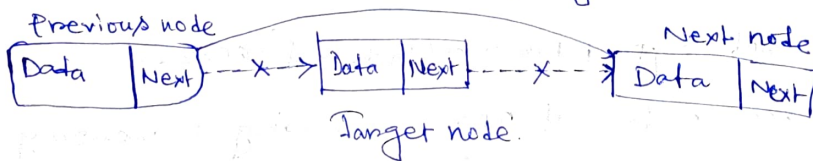
```
}
```

→ 10 → 20 → 30 → NULL

becomes 30 → 20 → 10 → NULL

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27) Next pointer Reassignment



Pointer adjustment,

```
prev->next = del->next;
```

```
free (del);
```

28) 5 → 8 → 12 → 3 → NULL

i) 7 → 5 → 8 → 12 → 3 → NULL

ii) 7 → 5 → 8 → 3 → NULL

iii) ~~7 → 5 → 8 → 12 → 3 → NULL~~ 7 → 5 → 8 → 3 → 9 → NULL

→ 7 → 5 → 8 → 3 → 9 → NULL

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